

Unit I – Basic Concepts of Data Structures & OOP

No.	Question	Marks	RBT	Source Paper(s)
1	Define Big - O Notation, Big Omega Notation, Big Theta Notation.	03	R	Summer-2025
2	Define linear data structure and give its examples.	03	R	Summer-2024
3	Define linked list. List different types of linked list.	03	R	Winter-2023
4	Explain Linear and Non Linear Data structure in Python with examples.	03	U	Winter-2025 (NEW)
5	Write names of linear data structures.	03	R	Winter-2024
6	Define Time and space complexity.	04	R	Winter-2024, Summer-2024
7	Enlist and Explain types of methods.	04	R	Winter-2025 (NEW)
8	Explain Constructor by giving suitable Example	04	U	Winter-2025 (OLD)
9	Explain Linear and Non Linear Data structure in Python with examples.	04	U	Winter-2023
10	1. Explain Inheritance. 2. Explain how to create a class and instance of class by giving example.	07	U	Winter-2025 (OLD)
11	1. Explain Linear and Non Linear Data Structure. 2. Develop a python code to explain concept of recursive function.	07	A	Winter-2025 (OLD)
12	Define Data Encapsulation & Polymorphism. Develop a Python code to explain Polymorphism.	07	A	Winter-2023
13	Design a student class for reading and displaying the student information, the getInfo() and displayInfo() methods will be used respectively. Where getInfo() will be a private method.	07	U	Summer-2025
14	Develop a class for managing student records with instance methods for adding and removing students from a class.	07	U	Winter-2024
15	Explain class, attributes, object and class method in python with suitable example.	07	U	Winter-2023
16	Explain concept of class & object with example.	07	U	Winter-2024
17	Explain instance method, class method and static method with example.	07	U	Summer-2024
18	Explain the concept of class and object with example.	07	U	Summer-2024
19	Write a Python class to represent a Cricketer. The class contains the name of the cricketer, team name and run as the data members. The member functions are as follows: to initialize the data members, to set run and display run.	07	U	Summer-2025

Unit II – Stack and Queues

No.	Question	Marks	RBT	Source Paper(s)
1	Define singly linked list.	03	R	Summer-2024
2	Differentiate between simple Queue and circular Queue.	03	U	Winter-2023
3	Differentiate between Stack and Queue.	03	U	Winter-2023, Winter-2025 (OLD), Summer-2025
4	Explain concept of recursive function.	03	U	Summer-2024
5	Explain DEQUEUE operation.	03	U	Winter-2025 (OLD)
6	Explain POP operation.	03	U	Winter-2025 (OLD)
7	Explain the importance of constructor in class.	03	U	Winter-2024
8	List applications of Stack and Queue.	03	R	Summer-2025
9	Write difference between circular and simple Queue.	03	U	Winter-2025 (NEW)
10	Write difference between stack and queue.	03	U	Winter-2024, Winter-2025 (NEW)
11	Convert following algebraic expression into postfix notation using Stack: i) $(a*b)*(c^{d+e}-f)$ ii) $a-b/(c*d/e)$	04	A	Summer-2025
12	Define recursion. Explain with example.	04	R	Summer-2025
13	Define stack and queue.	04	R	Summer-2024
14	Explain concept of circular queue.	04	U	Winter-2024
15	Explain concept of recursive function with suitable example.	04	U	Winter-2023
16	Explain different operations on stack.	04	U	Winter-2024
17	Explain Enqueue and Dequeue operations on Queue.	04	U	Summer-2024
18	Explain expression conversion from infix to postfix.	04	U	Winter-2024
19	Explain Fibonacci series using recursive function in python.	04	U	Winter-2025 (NEW)
20	Explain GCD using recursive function in python.	04	U	Winter-2025 (NEW)
21	Write an algorithm for PUSH and POP operation of stack in python.	04	A	Winter-2023
22	1. Differentiate between simple Queue and circular Queue. 2. Develop a python	07	A	Winter-2025 (OLD)

	code to implement Enqueue operation in Queue.			
23	1. Write an algorithm for PUSH operation on Stack. 2. Convert following equation from infix to postfix using Stack: $(A+B)*(C+D)$	07	A	Winter-2025 (OLD)
24	Consider the size of the stack as 5. Apply the following operation on stack and show status and top pointer after each operation: Push a,b,c; pop	07	A	Summer-2025
25	Convert expression $A+B/C+D$ to postfix and evaluate postfix expression using stack assuming some values for A, B, C and D.	07	A	Summer-2024
26	Convert following equation from infix to postfix using Stack: $A * (B + C) - D / (E + F)$	07	A	Winter-2023
27	Define stack? Given equation to conversion from infix to postfix using stack: $A*(C+D)+(J+K)*N+M*P$	07	R	Winter-2025 (NEW)
28	Describe evaluation algorithm of postfix expression $ABC+*D/$.	07	U	Winter-2024
29	Describe the procedure for inserting a new node after and before a given node in a singly linked list.	07	U	Winter-2024
30	Develop a program to implement a queue using a list that performs the following operations: enqueue, dequeue.	07	A	Summer-2025
31	Develop a python code to implement Enqueue and Dequeue operation in Queue.	07	A	Winter-2023
32	Explain and implement code for Enqueue and Dequeue Operation in Queue.	07	A	Winter-2025 (NEW)
33	Explain and implement code for POP and PUSH Operation in stack.	07	A	Winter-2025 (NEW)
34	Explain basic operations on stack.	07	U	Summer-2024
35	Write an algorithm for Evaluation of post fix expression. Also, Evaluate Post fix expression: 15, 6, 3, -, /, 2, 2, 6, +, *, +	07	A	Winter-2025 (NEW)

Unit III – Linked List

No.	Question	Marks	RBT	Source Paper(s)
1	Differentiate between circular linked list and singly linked list.	03	R	Summer-2025
2	Enlist applications of Linked List.	03	R	Summer-2024
3	Explain concept of doubly linked list.	03	U	Summer-2024, Winter-2025 (OLD)
4	Explain concept of Singly linked list.	03	U	Winter-2025 (OLD)
5	Explain searching an element in linked list.	03	U	Winter-2024
6	Explain traversing a linked list.	03	U	Winter-2024
7	Give Difference between Singly linked list and Circular linked list.	03	R	Winter-2023
8	List different operations performed on singly linked list.	03	R	Winter-2023
9	List types of linked lists. Give graphical representation of each type.	03	R	Summer-2025
10	Write a code to count the number of nodes in singly linked list.	03	A	Summer-2024
11	Write applications of linked list.	03	U	Winter-2024
12	Write difference between singly linked list and circular linked list.	03	R	Winter-2025 (NEW)
13	Write difference between singly linked list and doubly linked list.	03	R	Winter-2025 (NEW)
14	Define linked list. List different types of linked list.	04	R	Winter-2025 (OLD)
15	Differentiate between singly linked list and doubly linked list.	04	R	Winter-2024
16	Explain concept of Circular Doubly linked list.	04	U	Winter-2025 (OLD)
17	Explain concept of Circular linked list.	04	U	Winter-2023, Winter-2025 (OLD)
18	Explain concept of Doubly linked list.	04	U	Winter-2023
19	Explain concepts of circular linked lists.	04	U	Winter-2024
20	Explain creation of singly linked list in python.	04	R	Summer-2024
21	Explain three applications of linked list in brief.	04	R	Summer-2025
22	List different operations performed on singly linked list.	04	R	Winter-2025 (OLD)
23	Match appropriate options from column A and B: Singly Linked List; Doubly Linked List; Circular Linked List; Node in a Linked List.	04	U	Summer-2024
24	Write an algorithm to insert a node after a given node in the linked list.	04	A	Winter-2025 (NEW)
25	Write an algorithm to search a given node in a singly link list.	04	A	Summer-2025
26	Write an algorithm to search a node before a given node in the linked list	04	A	Winter-2025 (NEW)
27	Explain algorithm and Implement code to delete node from beginning of singly linked list.	07	A	Winter-2025 (NEW)
28	Explain deletion of first and last node in singly linked list.	07	U	Summer-2024
29	Implement a code To create and insert a node at the beginning of the singly linked list and display the list.	07	A	Winter-2025 (NEW)
30	Implement a program to create and display circular linked lists.	07	A	Summer-2025
31	Implement code To create and insert node at the end of the singly linked list and	07	A	Winter-2025 (NEW)

	display the list.			
32	Implement program to perform the following operation on singly linked list: 1) Insert a node at the beginning of a singly linked list. 2) Delete a node from the beginning of a singly linked list.	07	A	Summer-2025
33	List application of linked list. Write an algorithm to count the number of nodes in singly linked list.	07	A	Winter-2023, Winter-2025 (NEW)
34	Write a code to insert a new node at the beginning and end of singly linked list.	07	A	Summer-2024
35	Write a program to delete a node at the beginning and end of singly linked list.	07	A	Winter-2024
36	Write a python code to insert a new node at the beginning of the singly linked list.	07	U	Winter-2025 (OLD)
37	Write an algorithm for the following operation on singly linked list: (1) To insert a node at the beginning of the list. (2) To insert the node at the end of the list.	07	A	Winter-2023
38	Write an algorithm to count the number of nodes in singly linked list.	07	A	Winter-2025 (OLD)

Unit IV – Searching and Sorting

No.	Question	Marks	RBT	Source Paper(s)
1	Compare Linear search with Binary search.	03	U	Winter-2023, Winter-2025 (OLD)
2	Define Sorting and List out different sorting methods.	03	R	Winter-2025 (NEW)
3	Define sorting. List different sorting methods.	03	R	Winter-2023, Winter-2025 (OLD)
4	Explain bubble sort algorithm.	03	U	Winter-2024
5	Explain Bubble sort with example.	03	U	Winter-2025 (NEW)
6	Explain concept of selection sort algorithm.	03	U	Summer-2024
7	Explain linear search with example.	03	U	Winter-2025 (NEW)
8	Explain Merge sort with example.	03	U	Winter-2025 (NEW)
9	Write a program for Selection Sort Method.	03	A	Summer-2025
10	Write a program of Insertion Sort Method.	03	A	Summer-2025
11	Apply Insertion sort to the following data to arrange them in ascending order: 25 15 35 20 30 5 10	04	A	Summer-2025
12	Apply Quick Sort to the following data and arrange them in the proper manner: 5 6 1 8 2 9 10 15 7 13	04	A	Summer-2025
13	Differentiate Linear & Binary search.	04	U	Winter-2024
14	Explain bubble sort method.	04	U	Summer-2024
15	Explain concept of linear search.	04	U	Summer-2024
16	Write algorithm for Quick Sort with example.	04	U	Winter-2025 (NEW)
17	Write algorithm for selection sort with example.	04	U	Winter-2025 (NEW)
18	Write an algorithm for Insertion sort method.	04	A	Winter-2023, Winter-2025 (OLD)
19	Write an algorithm for selection sort method.	04	A	Winter-2023, Winter-2025 (OLD)
20	Develop a python code to sort the following list in ascending order using Bubble sort method: list1=[5,4,3,2,1,0]	07	A	Winter-2023
21	Develop a python code to sort the following list in ascending order using selection sort method: list1=[6,3,25,8,-1,55,0]	07	A	Winter-2023
22	Explain algorithm to search a particular element from list using Binary Search.	07	R	Winter-2024
23	Explain Bubble sort method with suitable example and write an algorithm for it.	07	A	Winter-2025 (OLD)
24	Explain Linear Search method with suitable example and write an algorithm for it.	07	A	Winter-2025 (OLD)
25	Explain Quick sort & Merge sort algorithm.	07	U	Winter-2024
26	Explain the working of quick sort method with example.	07	U	Summer-2024
27	Implement a python program to search a particular element from a list using Linear Search.	07	A	Summer-2025
28	Implement Merge sort algorithm.	07	A	Summer-2025
29	Write a code to implement binary search algorithm.	07	A	Summer-2024
30	Write a program to sort numbers in ascending order using selection sort algorithm.	07	A	Winter-2024

Unit V – Trees

No.	Question	Marks	RBT	Source Paper(s)
1	Define a complete binary tree.	03	R	Winter-2024
2	Define and give examples: Strict Binary Tree and Complete Binary Tree.	03	R	Summer-2025
3	Define the following terms regarding Tree data structure: (1) Forest (2) Root node	03	R	Winter-2023

	(3) Leaf node			
4	Define the following terms regarding Tree data structure: (1) In-degree (2) Out-degree (3) Depth	03	R	Winter-2023
5	Enlist applications of binary tree.	03	R	Summer-2024
6	Explain binary tree.	03	U	Summer-2024
7	Explain the following terms with suitable example: 1. Forest 2. Leaf node 3. Root node	03	U	Winter-2025 (OLD)
8	List Tree traversal methods. Explain any one.	03	R	Winter-2025 (OLD)
9	State the fundamental characteristic of a binary search tree.	03	R	Winter-2024
10	Write Short note on: Applications of Tree.	03	R	Summer-2025
11	Define Forest, Root Node and Leaf Node	04	R	Winter-2025 (NEW)
12	Define the terms root, path, parent and children with reference to tree.	04	R	Summer-2024
13	Define tree. List four applications of tree.	04	R	Winter-2025 (NEW)
14	Draw Binary search tree for 78,58,82,15,66,80,99 and write In-order traversal for the tree.	04	A	Winter-2023
15	Explain basic terminology of Binary Tree: Level number, Degree, In-degree, Out-degree, Leaf Node.	04	U	Summer-2025
16	Explain different Tree Traversal Methods.	04	U	Summer-2025
17	Explain inorder traversal of a binary tree.	04	U	Winter-2024
18	Explain insertion of a node in binary search tree.	04	U	Summer-2024
19	Explain postorder traversal of a binary tree.	04	U	Winter-2024
20	Explain the following terms with suitable example: 1. In-degree 2. Out-degree	04	U	Winter-2025 (OLD)
21	Write Preorder and postorder traversal of the following Binary tree.	04	U	Winter-2023, Winter-2025 (OLD)
22	Apply preorder and postorder traversal for given below tree.	07	A	Summer-2024
23	Construct a binary search tree for the following data items 25, 15, 20, 40, 10 (consider 25 as a root node). Delete 10 and 25 from this tree and reconstruct the tree.	07	A	Winter-2025 (NEW)
24	Construct Binary Search Tree for the following data: 5,3,1,4,8,7,10 (consider 5 as root). Write In-order traversal for it.	07	A	Winter-2025 (OLD)
25	Develop a program to implement construction of Binary Search Tree.	07	A	Winter-2023
26	Draw Binary search tree for 8, 4, 12, 2, 6, 10, 14, 1, 3, 5 and write In-order traversal for the tree.	07	A	Summer-2024
27	Show the construction of binary searching using nodes 7, 10, 9, 12, 13. Take 7 as a root node. Write post order and pre order traversal of created tree.	07	A	Winter-2025 (NEW)
28	Write a menu driven program to perform the following operation on Binary Search Tree: Create a BST.	07	A	Summer-2025
29	Write a menu driven program to perform the following operation on Binary Search Tree: Insert an element in BST.	07	A	Summer-2025
30	Write a program to delete a node from a binary search tree.	07	A	Winter-2024
31	Write a program to inserting a node into a binary search tree.	07	A	Winter-2024
32	Write an algorithm for the following operation: (1) Insertion of Node in Binary Tree (2) Deletion of Node in Binary Tree	07	A	Winter-2023
33	Write an algorithm for the following operation: i) Insertion of Node in Binary Tree ii) Deletion of Node in Binary Tree	07	A	Winter-2025 (OLD)